

1. We need to create a image interface(Image) and implement an entity of the image interface. Proxy image is a proxy class(ProxyImage) to reduce the memory consumption of real image object(RealImage) loading. Proxy pattern demo class(ProxyPatternDemo) use proxy image to get the image object to load and display as required. The image interface contains a display function. The real image class has a attribute of file name(fileName), load function(loadFromDisk) and display function(display). The proxy image class has a real image attribute(realImage), file name attribute(fileName) and a display function(display).

Image

```
package experiment6_1;

interface Image {

    void display();

}
```

ProxyImage

```
package experiment6_1;

public class ProxyImage implements Image{

    private RealImage realImage;

    private String fileName;

    public ProxyImage(String fileName) {

        this.fileName = fileName;

    }

    @Override
```

```

    public void display() {
        if(reallImage == null){
            reallImage = new ReallImage(fileName);
        }
        reallImage.display();
    }
}

```

ProxyPatternDemo

```

package experiment6_1;

public class ProxyPatternDemo {
    public static void main(String[] args) {
        String fileName = "test.png";
        ProxyImage proxyImage = new ProxyImage(fileName);
        proxyImage.display();
    }
}

```

ReallImage

```

package experiment6_1;

public class ReallImage implements Image{
    private String fileName;
}

```

```
public ReallImage(String fileName){  
    this.fileName = fileName;  
    loadFromDisk();  
}  
  
public void loadFromDisk(){  
    System.out.println("Loading image "+fileName+" from  
disk");  
}  
  
@Override  
public void display() {  
    System.out.println("Display image: "+fileName);  
}  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\In  
Loading image test.png from disk  
Display image: test.png  
  
Process finished with exit code 0
```